MTH208

Timed Online Assignment – July Semester 2021

Advanced Linear Algebra

Tuesday, 23 November 2021 **1:00 pm – 3:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **SIX (6)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

Calculate the Jordan canonical form of the matrix

$$\begin{pmatrix} 3 & 0 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 \\ -1 & 0 & 4 & 0 & 0 \\ 1 & 0 & -1 & 4 & 0 \\ -1 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

(15 marks)

Question 2

Let

$$A = \begin{pmatrix} 1 & -3 \\ -4 & 3 \\ 3 & 4 \end{pmatrix}$$

and let $B = A^*A$, a positive semi-definite matrix.

(a) Prove that the eigenvalues of B are 35 and 25, and illustrate the property of normal operators by constructing a unitary matrix Q and a diagonal matrix D such that $Q^*BQ = D$.

(8 marks)

(b) Find a singular value decomposition of A.

(8 marks)

(c) Give a brief explanation why the matrix AA^* is singular, that is, not invertible.

(2 marks)

(d) Find two different unitary matrices U_1, U_2 and an upper-triangular matrix R with non-negative eigenvalues such that

$$AA^* = U_1R = U_2R.$$

(7 marks)

Question 3

Let $\{w_1, w_2, w_3\}$ be a linearly independent set of vectors in $\mathbb{R}^4$. Suppose that

$$\langle w_1, w_1\rangle = 9, \langle w_1, w_2\rangle = -5, \langle w_1, w_3\rangle = 2,$$
$$\langle w_2, w_2\rangle = 4, \langle w_2, w_3\rangle = -1, \langle w_3, w_3\rangle = 1,$$

where $\langle x, y\rangle$ denotes the standard inner product of vectors x, y.

Let $W = \text{span}\{w_1, w_2, w_3\}$ and define a linear operator $T : W \longrightarrow W$,

$$T(w_1) = w_1 + w_3$$
$$T(w_2) = 2w_1 - w_2 + 3w_3$$
$$T(w_3) = w_2 - 2w_3.$$

(a) Use the Gram-Schmidt process to determine an orthogonal basis $\mathcal{C} = \{v_1, v_2, v_3\}$ of W, expressing each of the vectors v_1, v_2, v_3 as a linear combination of w_1, w_2, w_3.

(6 marks)

(b) Compute $[T]_{\mathcal{C}}$, the matrix representation of T relative to the basis $\mathcal{C}$.

(4 marks)

(c) Let $S : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$ be the orthogonal projection operator of $\mathbb{R}^4$ onto W. Let the 4×4 matrix B denote the matrix representation of S relative to the standard basis of $\mathbb{R}^4$. State the rank and nullity of B, and find a suitable matrix

$$\begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}$$

such that

$$B = \sum_{i=1}^{3} \sum_{j=1}^{3} c_{ij} w_i {w_j}^T ,$$

where ${w_j}^T$ denotes the transpose of the column vector w_j.

(6 marks)

(d) If V is a nonzero subspace of $\mathbb{R}^n$ (where n is a positive integer) having an orthonormal basis of column vectors $\{x_1, x_2, x_3, \dots, x_m\}$, and $S : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ is the orthogonal projection operator of $\mathbb{R}^n$ onto V, show that the matrix representation of S relative to the standard basis of $\mathbb{R}^n$ is

$$\sum_{i=1}^{m} x_i {x_i}^T .$$

(4 marks)

Question 4

Let

$$A = \begin{pmatrix} 1 & 2 & -3 \\ 2 & 0 & 3 \\ -3 & 3 & -5 \end{pmatrix}$$

be the matrix representation of a symmetric bilinear form f on $\mathbb{R}^3$ relative to the standard basis. Find an invertible matrix Q such that QAQ^T is a diagonal matrix, and determine the rank and signature of f.

(10 marks)

Question 5

Let

$$w_1 = \begin{pmatrix} 1 \\ -3 \\ -2 \\ 1 \end{pmatrix}, \qquad w_2 = \begin{pmatrix} 3 \\ 0 \\ -2 \\ 1 \end{pmatrix}, \qquad w_3 = \begin{pmatrix} 2 \\ 2 \\ 4 \\ 2 \end{pmatrix}, \qquad w_4 = \begin{pmatrix} 2 \\ 3 \\ 2 \\ 1 \end{pmatrix}.$$

(a) It is known that $\mathcal{B} = \{w_1, w_2, w_3, w_4\}$ is a basis of $V = \mathbb{R}^4$. Find the dual basis $\mathcal{B}^*$ of V^*.

(4 marks)

(b) Let W be the subspace of $\mathbb{R}^4$ spanned by $\{w_1, w_2\}$. Find a basis for the annihilator W^0.

(4 marks)

(c) The linear functionals h_1, h_2, h_3 on $V = \mathbb{R}^4$ are defined as follows:

$$h_1(a, b, c, d) = a + c + 2d,$$
$$h_2(a, b, c, d) = a + b,$$
$$h_3(a, b, c, d) = 3a + 4c,$$

for any $a, b, c, d \in \mathbb{R}$. Let H denote the subspace of V^* spanned by h_1, h_2, h_3. Show that $\dim(H) = 3$ and find a basis of H^* that is dual to $\{h_1, h_2, h_3\}$.

(5 marks)

(d) Let V be a finite dimensional vector space over $\mathbb{R}$. If $\phi : V \longrightarrow V$ is a linear operator on V, we can define a linear operator $\phi^* : V^* \longrightarrow V^*$ on the dual space V^* as follows:

$$\phi^*(T) = T \circ \phi \quad \text{for all } T \in V^* .$$

Show that if $\mathcal{B}$ is an ordered basis of V and A is the matrix representation of ϕ with respect to $\mathcal{B}$, then A^T (the transpose of A) is the matrix representation of ϕ^* with respect to dual basis $\mathcal{B}^*$.

(7 marks)

Question 6

Let V be a complex inner product space and let $\{ v_1, v_2, \dots, v_n \}$ be an orthonormal basis of V. Suppose that $\{ w_1, w_2, \dots, w_n \}$ is a set of non-zero vectors in V such that

$$\|w_i - v_i\| < \frac{1}{\sqrt{n}},$$

for all i, $1 \leq i \leq n$. Show that $\{ w_1, w_2, \dots, w_n \}$ is a basis of V.

(10 marks)

----- END OF PAPER -----